Angles of inclination

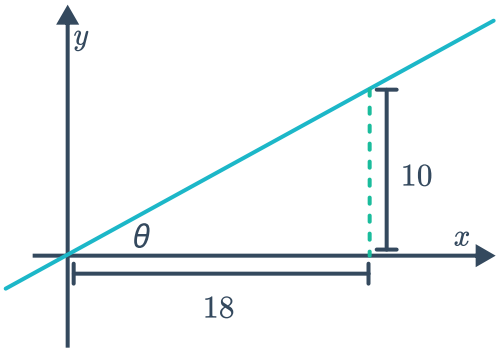
1 Find the slope of a line that is inclined at an angle of 50° to the positive x-axis. Round your answer to one decimal place.

2 Find the measure of the angle of inclination of a line to the positive x-axis. whose slope is 4. Round your answer to the nearest degree.

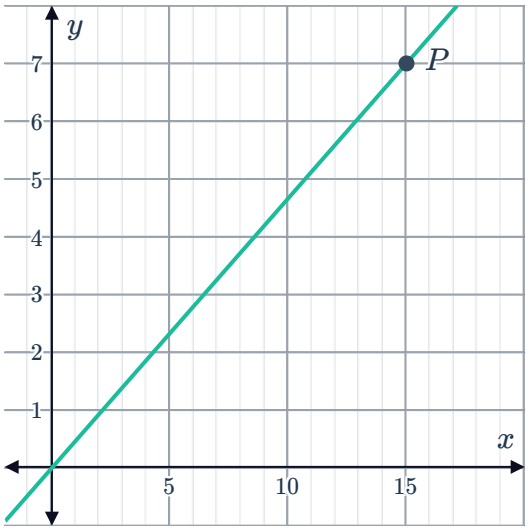
3 Consider the given line with an angle of inclination of θ with the positive x-axis:

a Find the slope of the line.

b Now, find the measure of θ. Round your answer to two decimal places.

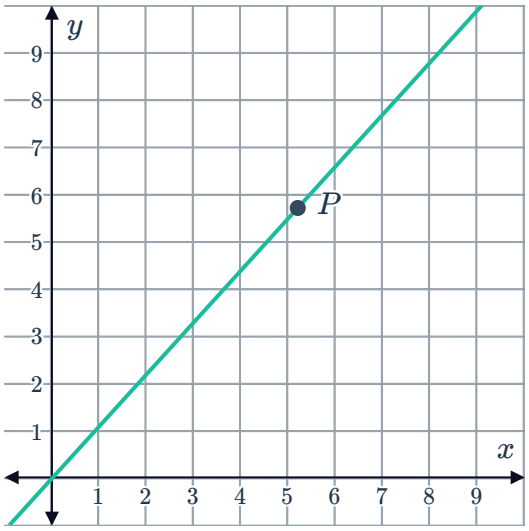


4 In the following graph, point $P\,(15, 7)$ lies on the line passing through the origin. Find the measure of θ, the angle of inclination of the line with the positive x-axis. Round your answer to two decimal places.



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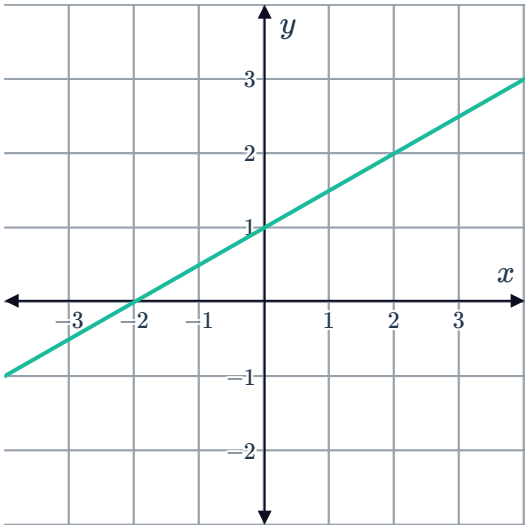
5 In the following graph, point $P\,(5.2, 5.72)$ lies on the line passing through the origin. Find the measure of θ, the angle of inclination of the line with the positive x-axis. Round your answer to two decimal places.



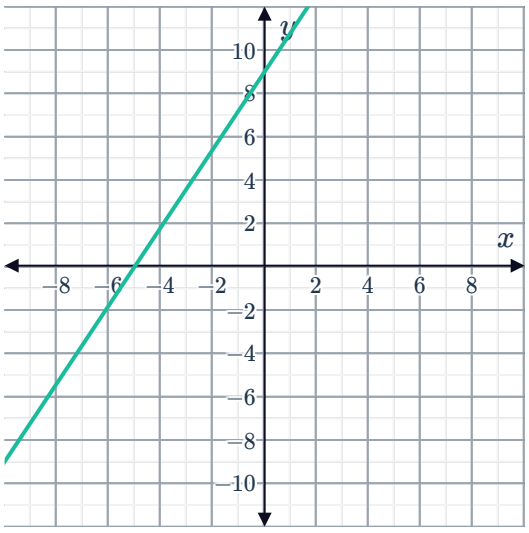
6 Consider the graph of the line $y = \frac{1}{2}x + 1$ given and the triangle whose vertices are the origin, the x-intercept and the y-intercept.
θ is the acute angle in the triangle that the line makes with the x-axis.

a State the slope of the line.

b Find the value of $\tan\theta$.



7 Consider the following graph:
Find the measure of θ, the angle of inclination of the line with the positive x-axis, if the x-intercept is -5 and the y-intercept is 9. Round your answer to two decimal places.



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8 For each of the following linear equations, find the measure of θ, the angle of inclination of the line with the positive x-axis. Round your answer to the nearest degree.

a $y = 6x + 2$

b $5x + 6y - 2 = 0$

9 A line makes an angle of θ with the positive x-axis. If $\theta < 90^\circ$. State whether the slope of the line is positive or negative.

10 Find the measure of θ, the angle of inclination of the line with the positive x-axis for a line with a slope of $-\dfrac{1}{\sqrt{3}}$.

11 A line is drawn on the coordinate plane at an angle of inclination θ with the positive x-axis. Given that $\sin\theta = \dfrac{12}{13}$, find the slope m of the line.

12 If a line has a slope of -4 and an angle of inclination of θ with the positive x-axis. State the interval in which the measure of θ could lie in. Write your answer as an inequality using degrees.

13 Two points $A\,(1, -2)$ and $B\,(9, 30)$ lie on a line that makes an angle of θ degrees with the positive x-axis.

a Find the slope of the line.

b Find the measure of θ to two decimal places.

14 A line with slope m has an angle of inclination of 65° with the positive x-axis such that $\tan 65^\circ = m$. Without solving for m, find the measure of θ such that $\tan\theta = -\dfrac{1}{m}$.

15 A horizontal line makes an angle of 0° with the x-axis.

a Evaluate $\tan 0^\circ$ by considering its relationship to the slope of the line.

b A vertical line is perpendicular to a horizontal line. Using the fact that the slopes of perpendicular lines have the relationship $m_1 \cdot m_2 = -1$, find the value of $\tan 90^\circ$.

16 Line L_1 makes an angle of θ with the positive x-axis. Line L_2 is such that it is perpendicular to line L_1. Form an expression for the slope of line L_2 in terms of θ.

17 A line passing through the points $(2, -3)$ and $(x, 9)$ makes an angle of 120° with the positive x-axis. Find the exact value of x.

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